Demo Client · DEMO-0013
13 November 2026 · v1.0

Field	Value
Document	Methodology of the AI automation and agent evaluation
Template version	ai_automation 1.0.0 / proquant-core 0.1.0
Engagement	DEMO-0013 · Demo Client
Process automated	Customer email triage and replies
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1. Purpose

To set out each step that turns the client's files into the scores, measures and verdict of the engagement, with every parameter, so that a second analyst can repeat the run on the same files and reach the same numbers.

2. Inputs

Data set	Size	Fingerprint (SHA-256)
processes	14 rows	`4475fef3ed9135e4`
cases	1,200 rows	`134314163a811724`
labels	400 rows	`8b501f2aaa82c637`
pilot	2,411 rows	`bf80e280a96b3d72`

3. Scoring the processes

For each process: annual hours = cases a month × minutes × 12 / 60; annual cost = hours × hourly cost. The automatable share is 0.9 of the rule-based share plus, for the remainder, a share set by input type (text 0.60, document 0.50, structured 0.80, voice 0.30), capped at 0.95. Value a year = annual cost × automatable share. Feasibility is the mean of three scores in [0, 1]: how digital the input is (structured 1.0, text 0.9, document 0.6, voice 0.4), volume (cases a month / 500, capped at 1) and the rule-based share. Risk is 0.1, 0.4 or 0.8 for low, medium or high decision risk, plus 0.1 where personal data is involved. Verdicts: risk ≥ 0.8 → keep human; value at or above the median and feasibility ≥ 0.6 → quick win; value at or above the median and lower feasibility → big bet; lower value and feasible → easy, small; otherwise not now. Build cost is a number of days per verdict (quick win 6, big bet 15, easy 3, keep human 4) at the agreed day rate; running cost is cases × 12 × the estimated AI cost per case plus the platform fee; payback = build cost / monthly net value.

4. Preparation and privacy

Case references are trimmed and duplicates removed. Each message is masked before any engine reads it, by four regular expressions applied in order: e-mail address, IBAN, card number (13 to 19 digits with optional

separators), phone number (international prefix or a ten-digit local number starting with 0). Each match is replaced by its tag. A second, broader scan (an @ followed by a domain, a run of nine or more digits not preceded by an order prefix, four groups of four digits) counts what the masking may have missed. Labelled cases without a message are set aside and counted.

5. The agent

One request per message to the Anthropic Messages API: the masked message as the user turn; a system prompt listing every request type and the reply rules; structured output (`output_config.format`) holding the answer to a JSON schema (request type from a fixed list, confidence 0 to 1, order number, urgency, language, summary, reply draft); a configured effort level; automatic prompt caching; the server-side refusal fallback. Every answer is appended to a cassette file with its tokens, time and request id, so the evaluation can be replayed without new calls. A keyword engine with no AI runs on the same cases: for each request type, the count of its keywords found in the message; the type with most hits wins, with confidence = its share of all hits × 0.85; order numbers by the pattern `SO–` followed by six or seven digits.

6. Routing

The route is computed by fixed rules from the answer and the policy, never by the model: an engine error or refusal → escalate; a request type outside the list → route to a team; urgent → escalate; otherwise the type's route (auto, review, route, escalate), except that an automatic route with confidence below the agreed threshold becomes review. The team's label maps to an expected route: automatic reply → auto, draft for review → review, route to team → route, escalate → escalate.

7. Evaluation measures

Accuracy: share of labelled cases whose request type matches the label, with a 95% interval from 1,000 bootstrap resamples (seed 13). Per request type: precision, recall and F1. Order accuracy: among labels with an order number, the share copied exactly. Unsafe: cases routed auto whose expected route is not auto. Error in auto: among cases routed auto, the share with the wrong request type. Complaint recall: the share of labelled complaints read as complaints. Missed escalations: expected escalate, routed otherwise. Cost per case: input tokens × input price + output tokens × output price, per million tokens, at the list prices in the configuration. Latency: median and 95th percentile of the time to an answer. The threshold curve repeats routing and the auto measures for thresholds 0.50 to 0.95 in steps of 0.05, 0.97 and 0.99.

8. Business case

For the process automated: messages a month from the inventory; minutes by hand = median minutes in the message export; hours back = messages × (auto share × minutes + review share × (minutes – 1.5)) / 60, where 1.5 minutes is the time to check and approve a draft; value = hours × hourly cost; running cost = messages × cost per case + platform fee / 12; payback = build cost / (value – running cost).

9. The pilot

Weeks run Monday to Sunday. A case answered alone and sent gives back the minutes by hand; an approved or edited draft gives back those minutes less the review time logged; escalated, rejected and corrected cases give back nothing. Edit rate = (edited + rejected) / reviewed. Realised against plan = hours back per pilot case / hours back per case in the business case. Drift = population stability index of the request-type mix, first half of the pilot against the second, with shares floored at 0.0001.

10. The automation pack

Built from the same prompt, schema, request settings and policy as the evaluation. The n8n workflow has eight steps: trigger, masking (the same four patterns in JavaScript), the API request, the route decision (the same rules as section 6), a four-way switch, one placeholder per route and a log step. It is read back after it is written: every connection points to a node, and the request carries the evaluated schema and prompt. The golden set is the labelled cases after masking, with the expected route.

11. Assessment

Fifteen go-live tests compare a measured value with thresholds agreed at kickoff (`tests:` in the configuration). A test without its data is not run. Any FAIL gives "Not ready to go live", any WARN "Ready, with a person reviewing", otherwise "Ready to go live". Each test that does not pass carries one action, an owner role and a date 14 days (FAIL) or 30 days (WARN) after the run.

12. Reproducibility

The run folder holds the configuration hash, input fingerprints, code version, every table as CSV, the recorded model answers and the log. Replaying the recorded answers with the same configuration gives the same tables.

13. Limitations

The limits of this work are set out in the report. No deviations from the standard methodology.